LIFTING A MOD-PI^8 RESOLUTION TO A SOLUTION BY COMBINING THE LIFTING AND DESINGULARIZATION LEMMAS

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'We omit the verification that this is a solution to the problem posed by the lemma'; the omitted verification is exactly the assembly chase of step 3 and is carried as an explicit core obligation.

Editorial note. Input lemma-lifting is a published entry (tag 07CP) used as a standing input by its display number; not proven on the author's behalf.

Allowed standard prerequisites (editorial).

- Lifting lemma with annihilator hypotheses. lemma-lifting (same chapter, PUBLISHED 07CP). Produces D -> Lambda with the smoothness of R -> D at primes off pi and over the smooth locus of the mod-pi^8 resolution.
- Desingularization with compatible mod-pi^4 maps. lemma-desingularize-strictly-standard (carded L26808). Produces B with A -> B -> Lambda and D -> B -> Lambda and the inclusion H_{D/R}B inside H_{B/R}.

Source reference. Original lemma 07CT: sources/stacks-smoothing/pinned-original-source.tex, source lines 2029–2029. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CP: sources/stacks-smoothing/pinned-original-source.tex, source lines 1638–1638. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a Noetherian ring. Let Λ be an R-algebra. Let $\pi \in R$ and assume that $Ann_R(\pi) = Ann_R(\pi^2)$ and $Ann_\Lambda(\pi) = Ann_\Lambda(\pi^2)$. Let $A \to \Lambda$ be an R-algebra map with A of finite presentation and assume π is strictly standard in A over R. Let*

$$A/\pi^8 A \to \bar{C} \to \Lambda/\pi^8\Lambda$$

be a factorization with $\bar{C}$ of finite presentation. Then we can find a factorization $A \to B \to \Lambda$ with B of finite presentation such that $R_\pi \to B_\pi$ is smooth and such that

$$H_{\bar{C}/(R/\pi^8 R)} \cdot \Lambda/\pi^8\Lambda \subset \sqrt{H_{B/R}\Lambda} \bmod \pi^8\Lambda.$$

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Apply Lemma 07CP to get $R \to D \to \Lambda$ with a factorization $\bar{C}/\pi^4\bar{C} \to D/\pi^4 D \to \Lambda/\pi^4\Lambda$ such that $R \to D$ is smooth at any prime not containing π and at any prime lying over a prime of $\bar{C}/\pi^4\bar{C}$ where $R/\pi^8 R \to \bar{C}$ is smooth. By Lemma 07CT we can find a finitely presented R-algebra B and factorizations $A \to B \to \Lambda$ and $D \to B \to \Lambda$ such that $H_{D/R}B \subset H_{B/R}$. We omit the verification that this is a solution to the problem posed by the lemma. $\square$